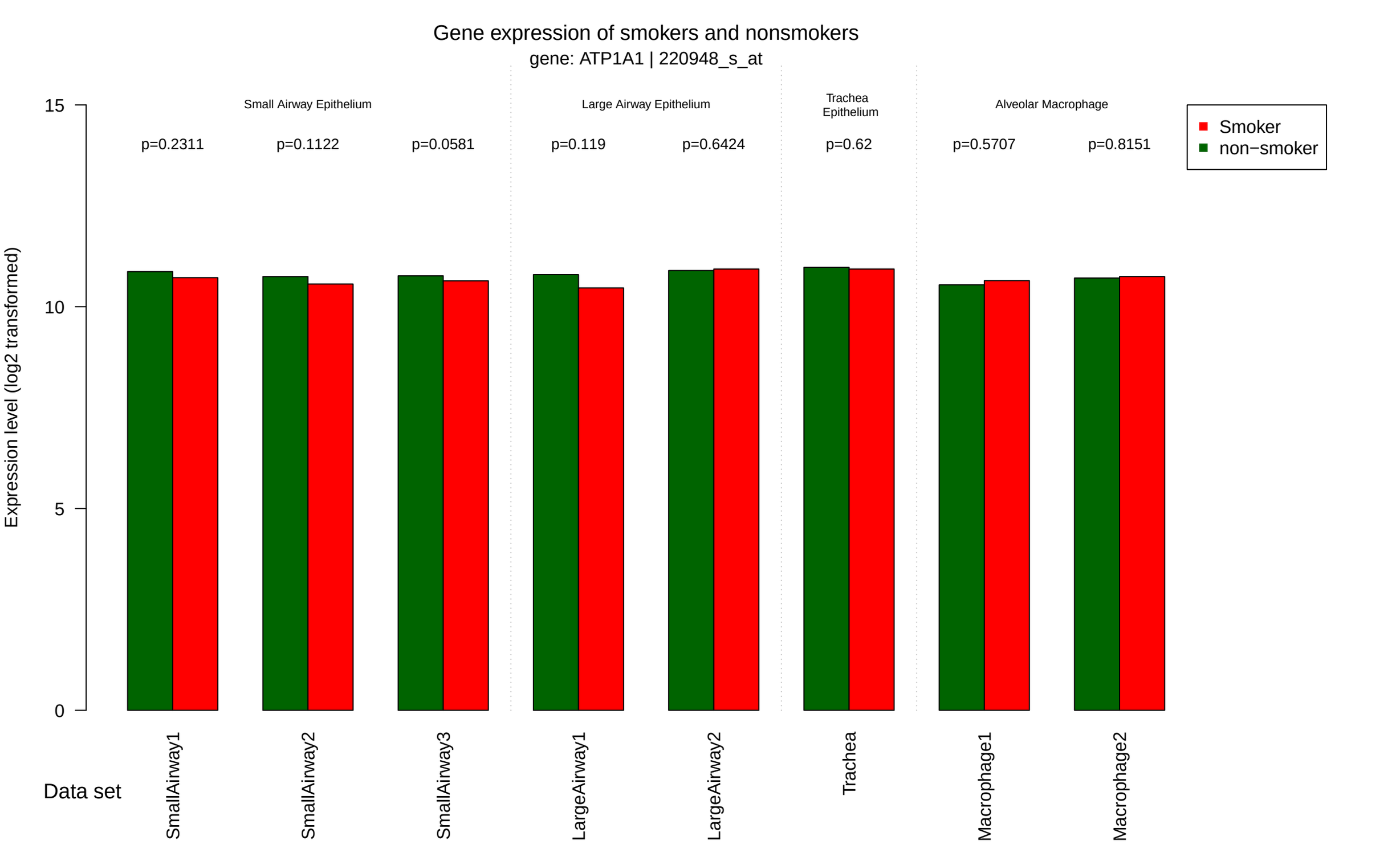
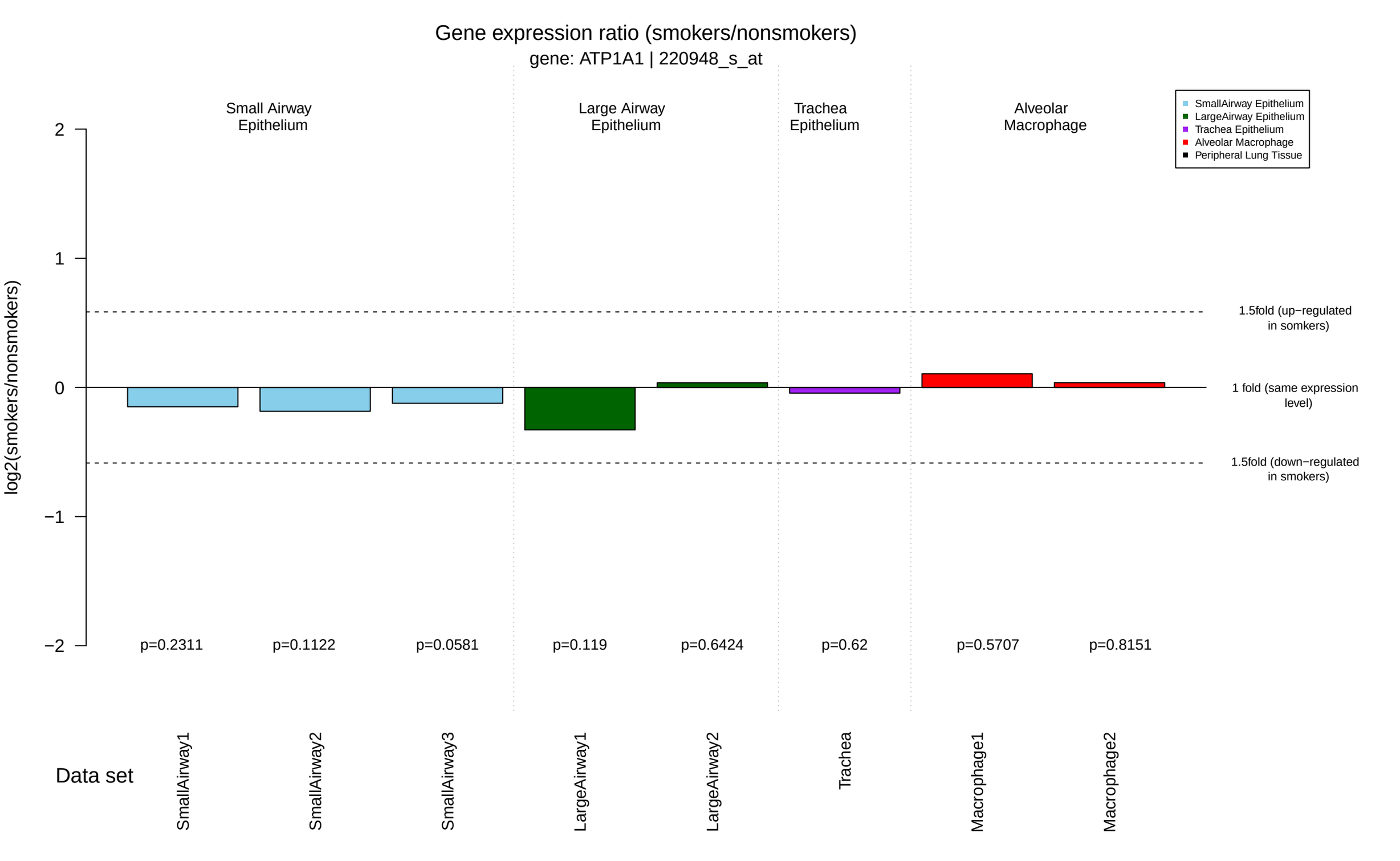
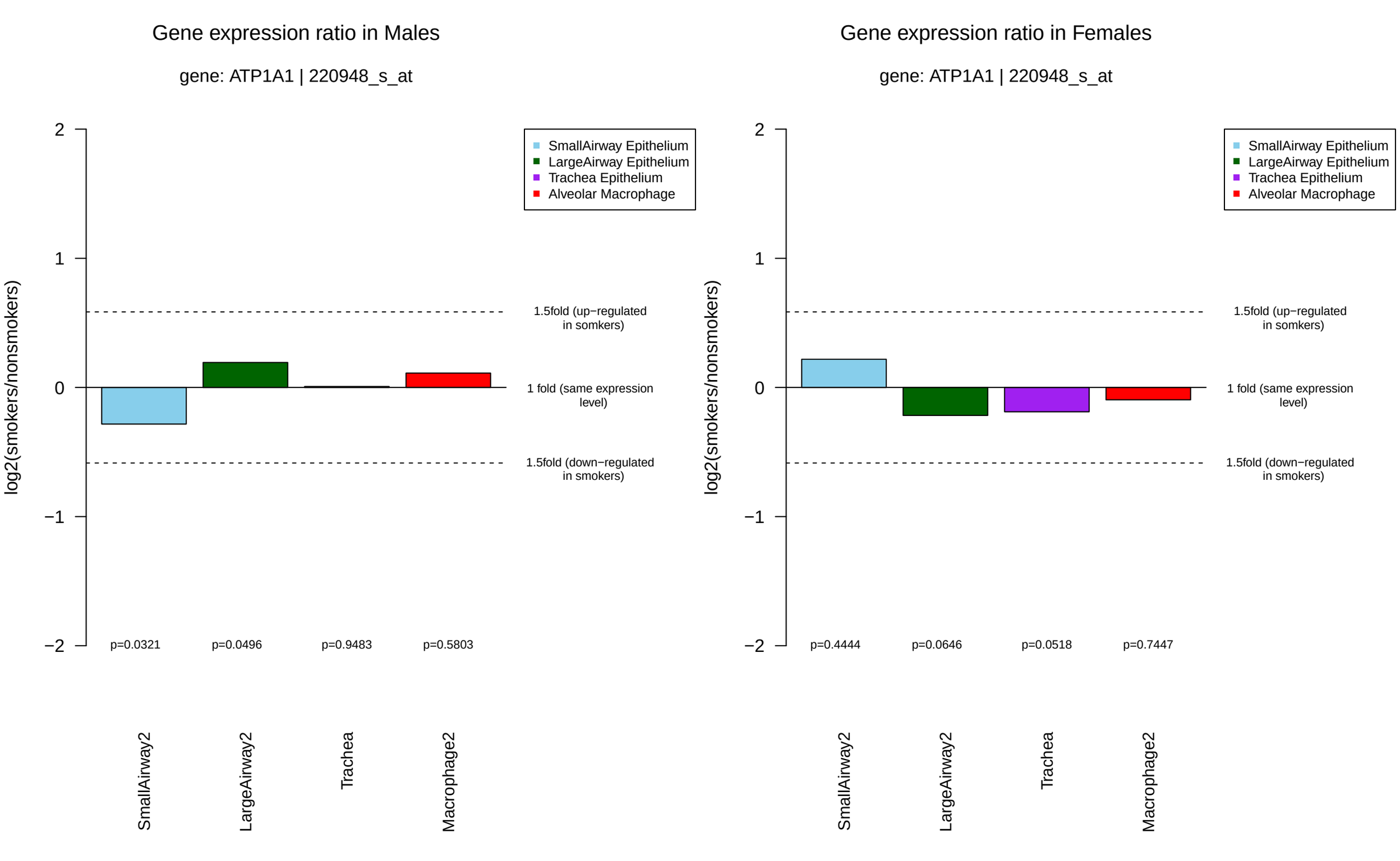
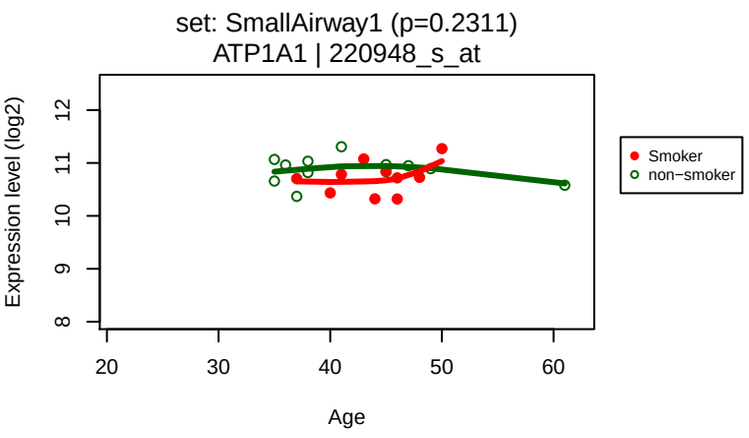
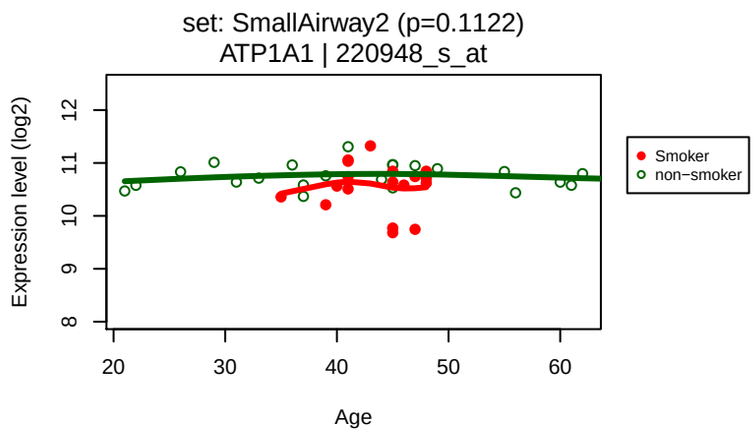
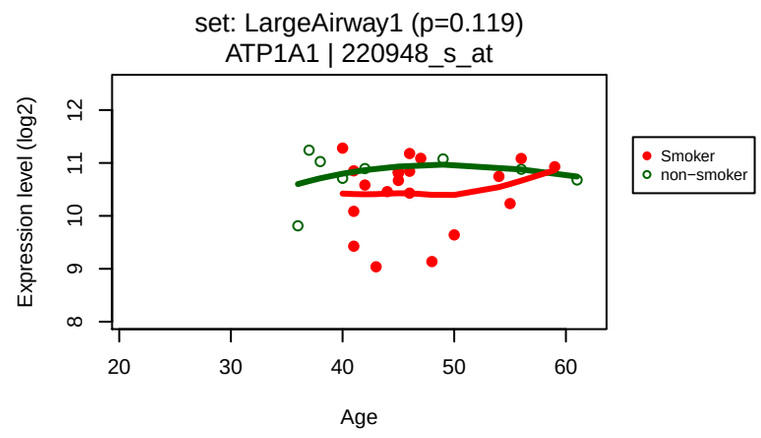

Gene expression of smokers and nonsmokers
gene: ATP1A1 | 220948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2311
p=0.1122
p=0.0581
p=0.119
p=0.6424
p=0.62
p=0.5707
p=0.8151
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

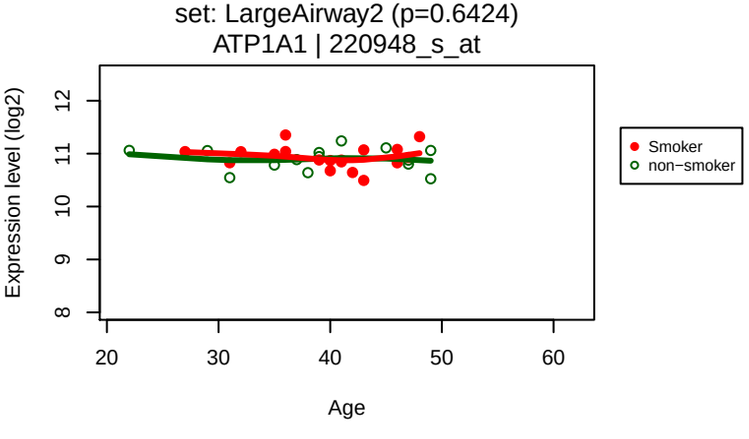

Gene expression ratio (smokers/nonsmokers)
gene: ATP1A1 | 220948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2311
p=0.1122
p=0.0581
p=0.119
p=0.6424
p=0.62
p=0.5707
p=0.8151
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

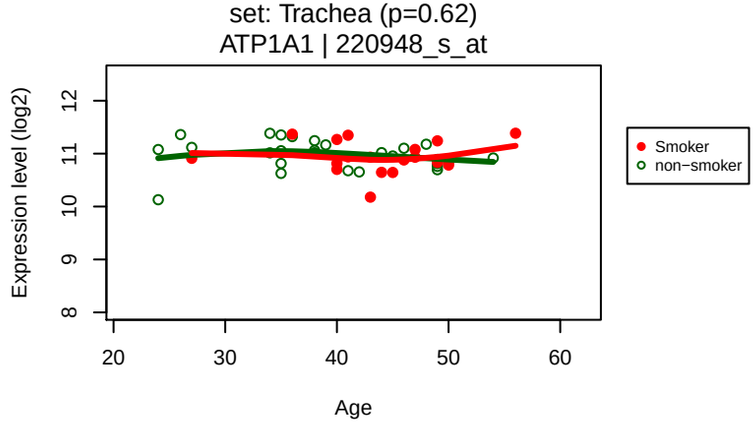

Gene expression ratio in Males
gene: ATP1A1 | 220948_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0321
p=0.0496
p=0.9483
p=0.5803
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP1A1 | 220948_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4444
p=0.0646
p=0.0518
p=0.7447
SmallAirway2
LargeAirway2
Trachea
Macrophage2

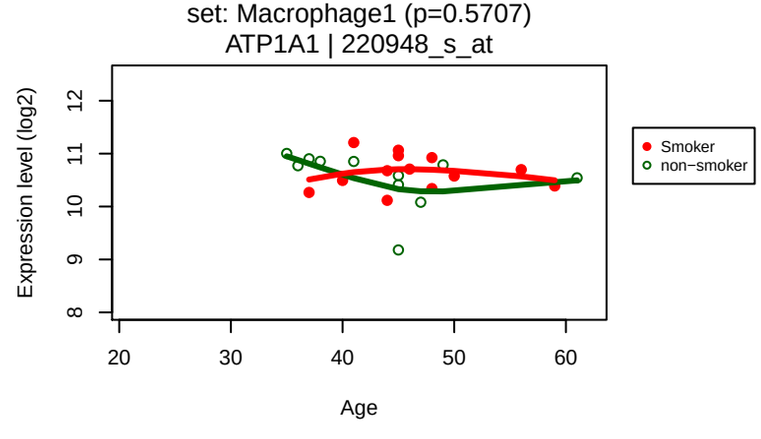

set: SmallAirway1 (p=0.2311)
ATP1A1 | 220948_s_at
Expression level (log2)
Age
Smoker
non−smoker

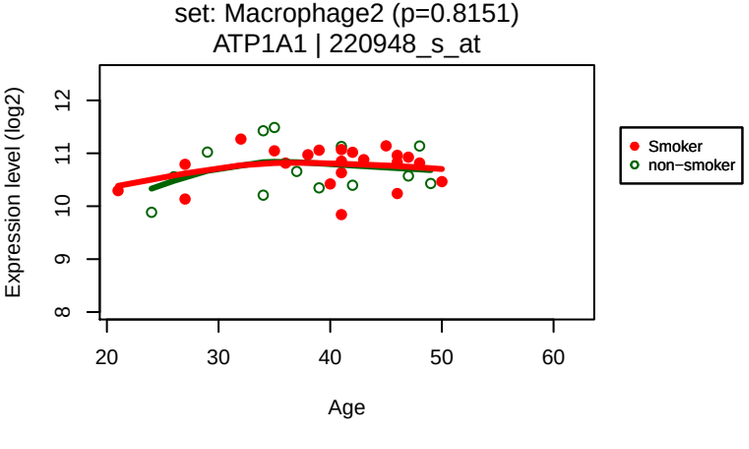

set: SmallAirway2 (p=0.1122)
ATP1A1 | 220948_s_at
Expression level (log2)
Age
Smoker
non−smoker

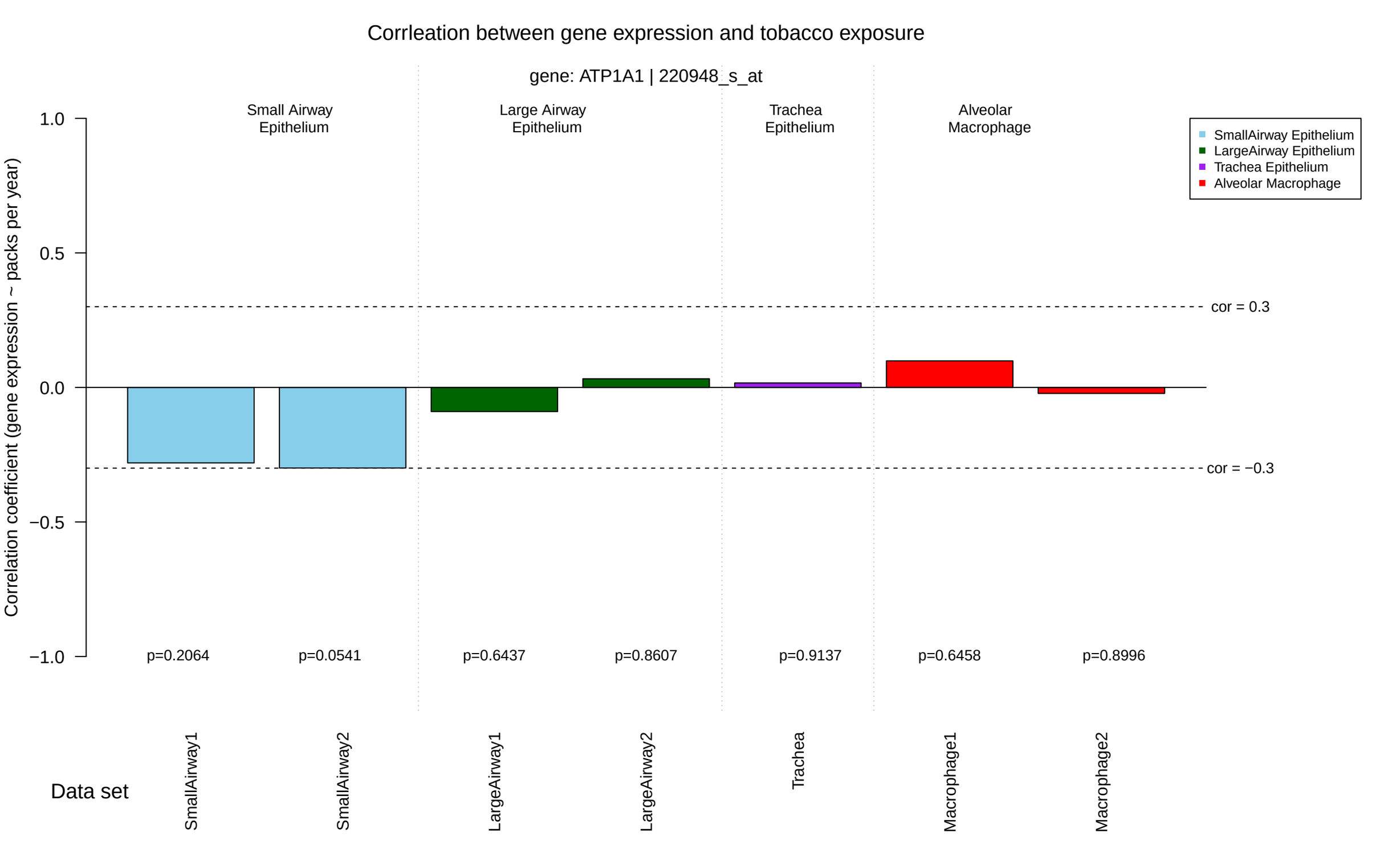

set: LargeAirway1 (p=0.119)
ATP1A1 | 220948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6424)
ATP1A1 | 220948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.62)
ATP1A1 | 220948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5707)
ATP1A1 | 220948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8151)
ATP1A1 | 220948_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP1A1 | 220948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2064
p=0.0541
p=0.6437
p=0.8607
p=0.9137
p=0.6458
p=0.8996
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2